

Chemical Process Industries Roadmap

CHE261

1 Introduction to the Chemical Process Industry

1.1 Structure and Classification

- **Industry structure:** raw materials (~ 10 ; oil, gas, coal, sulfur, air, water) $\rightarrow$ fuels (~ 10) $\rightarrow$ base chemicals (~ 20 ; ethylene, ammonia, methanol, sulfuric acid, chlorine) $\rightarrow$ intermediates (~ 300) $\rightarrow$ final products ($\sim 30,000$).
- **Bulk chemicals:** large-volume, low unit cost, economy of scale (e.g. H_2SO_4).
- **Fine chemicals:** smaller-scale, batch production, higher unit cost (e.g. APIs).
- **Specialty chemicals:** formulated products for specific end uses (coatings, adhesives, pesticides).
- Key historical milestones: lead-chamber H_2SO_4 (1746), Leblanc/Solvay soda ash, Haber-Bosch ammonia (1905), Fischer-Tropsch synfuels (1923), zeolite catalytic cracking (1964).

1.2 General Considerations in Process Design

Any industrial synthesis route is evaluated on: yield, conversion and selectivity (mass balances); energy usage; reaction kinetics and productivity; number of reaction/separation steps; scale and manufacturing cost; environmental impact (waste, emissions, effluent); health and process safety; and raw-material availability.

1.3 Process Safety: The Bhopal Gas Tragedy

- On the night of 2–3 December 1984, methyl isocyanate (MIC), an intermediate in the Sevin (carbaryl) pesticide process, leaked from a Union Carbide plant in Bhopal, causing history's worst industrial accident (thousands of deaths, hundreds of thousands injured).
- Contributing factors: a critical isolation valve left open, a disabled flare tower and vent-gas scrubber, an undersized water curtain, inoperable refrigeration (MIC must be kept cold), and inadequate staffing, maintenance and emergency planning.
- Lessons: minimise inventory of hazardous intermediates, keep safety systems operational and tested, maintain buffer zones from residential areas, and have emergency response plans in place. A related case is the 1974 Flixborough (UK) explosion, caused by a temporary bypass pipe in a cyclohexane oxidation plant.

2 Chemical Reactor Fundamentals

The general mole balance on species A over a control volume is:

$$F_{A0} - F_A + \int_V r_A dV = \frac{dN_A}{dt}$$

- **Batch reactor:** no flow in/out, perfectly mixed; $\frac{dC_A}{dt} = r_A$ (constant volume). Used for small-scale, versatile production.
- **CSTR:** continuous feed/removal, perfectly mixed, uniform composition equal to the exit stream; at steady state $F_{A0} - F_A + r_A V = 0$. Good temperature control; used for liquid-phase reactions.
- **PFR/PBR:** tubular, no axial mixing (ideal), composition varies along length; $\frac{dF_A}{dV} = r_A$. Used for gas-phase and heterogeneous catalytic reactions (PBR: rate based on catalyst mass).
- **Continuous operation** generally gives constant reactant/product concentrations, low inventory of hazardous material, better containment, automated control, and reduced waste versus batch operation.
- Flowsheets are represented at increasing levels of detail: block diagrams, simplified engineering flow sheets (with mass/energy balance data), and P&ID (piping and instrumentation diagrams).

3 Petroleum Refining

3.1 Crude Distillation

Crude oil is heated ($\sim 315\text{--}370^\circ\text{C}$) and separated in an atmospheric tower into wet gas, naphtha, kerosene, gas oil and topped (reduced) crude; the reduced crude is further separated under vacuum into vacuum gas oil and vacuum residue.

3.2 Fluid Catalytic Cracking (FCC)

- Breaks heavy gas oils into lighter products (LPG, gasoline, middle distillates) using heat and an acid catalyst, without added hydrogen.
- Catalyst (zeolite Y in a silica–alumina matrix) is continuously circulated between the riser **reactor** ($\sim 90\%$ conversion in 2–5 s residence time, $490\text{--}570^\circ\text{C}$) and the **regenerator** ($\sim 700^\circ\text{C}$), where coke is burnt off in air.
- Key variables: reactor temperature, catalyst-to-oil ratio, and feed preheat; increasing temperature raises conversion but gasoline yield passes through a maximum due to secondary cracking.
- Additives: ZSM-5 boosts octane/light olefins; Pt-based combustion promoters and SO_x /metal-passivation additives control emissions and catalyst poisoning.

3.3 Catalytic Reforming and Isomerization

- Converts low-octane naphtha into high-octane reformat via simultaneous dehydrogenation of naphthenes to aromatics, isomerization, dehydrocyclization (paraffins to aromatics) and (undesired) hydrocracking/dealkylation.
- Dual-functional catalyst: a metal site (Pt, or Pt–Re) for (de)hydrogenation and an acid site (chlorided alumina) for isomerization/cyclization.
- Operated at $455\text{--}510^\circ\text{C}$, 10–35 atm; a compromise between high reaction rates and coke formation (favoured by high T, low P).
- **Isomerization** of light n-C₅/n-C₆ to branched isomers (higher octane) uses a similar dual-functional Pt/chlorided-alumina catalyst at milder conditions ($120\text{--}175^\circ\text{C}$).

3.4 Alkylation

Combines an iso-paraffin (i-C₄) with a light olefin to form branched, high-octane iso-paraffins, e.g. $\text{iso-C}_4\text{H}_{10} + \text{C}_4\text{H}_8 \longrightarrow \text{iso-C}_8\text{H}_{18}$. Highly exothermic; catalyzed by concentrated H_2SO_4 ($10\text{--}20^\circ\text{C}$) or HF ($\sim 40^\circ\text{C}$) via a carbocation mechanism.

3.5 Hydrocracking and Hydrotreating

- **Hydrocracking**: cracking with added H_2 over a catalyst at moderate T and high P, converting heavy, aromatic or high-S/N feedstocks into naphtha, jet and diesel fuel.
- **Hydrotreating**: uses H_2 to remove sulfur (hydrodesulfurization) and nitrogen as H_2S and NH_3 , and saturates olefins, upgrading feed prior to cracking/reforming.
- Increasing temperature accelerates cracking faster than desulfurization/denitrogenation; higher pressure and H_2 recycle reduce coking and favour heteroatom removal.

3.6 Delayed Coking

Severe thermal cracking of vacuum residue in large coke drums (furnace outlet $\sim 500^\circ\text{C}$, drum $\sim 1.4\text{--}2.1$ atm), producing coke, naphtha, gas oil and gas; used to eliminate low-value residual fuel oil. Coke yield increases with drum pressure and decreases with drum temperature.

4 Inorganic Heavy Chemicals

4.1 Sulfuric Acid: The Contact Process

- Chemistry: $\text{S} + \text{O}_2 \longrightarrow \text{SO}_2$ then $\text{SO}_2 + \frac{1}{2} \text{O}_2 \rightleftharpoons \text{SO}_3$ (exothermic, reversible), catalyzed by V_2O_5 on silica ($410\text{--}440^\circ\text{C}$) or Cs-promoted catalyst ($360\text{--}400^\circ\text{C}$).
- Steps: (1) burn S to SO_2 , (2) cool, (3) catalytic oxidation to SO_3 in a multi-stage converter with interstage cooling, (4) cool, (5) absorb SO_3 in 98% H_2SO_4 .

- **Double absorption** (a second SO_3 absorber) raises conversion from $\sim 98\%$ to $>99.5\%$, reducing SO_2 emissions; it is the industry standard.
- Oleum (fuming H_2SO_4) is made by absorbing excess SO_3 into H_2SO_4 . Largest end use worldwide is fertilizer manufacture.

4.2 Ammonia: The Haber–Bosch Process

- Chemistry: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, exothermic ($\Delta H \approx -109 \text{ kJ/mol}$ at 500°C); equilibrium favours high P and low T, but rate requires higher T, so an active catalyst (K-promoted Fe, or Ru/C) is essential.
- Typical conditions: 150–300 atm, $350\text{--}400^\circ\text{C}$ feed; multistage adiabatic reactors with interstage (quench) cooling give a declining optimum temperature profile.
- Single-pass conversion is low ($\sim 15\text{--}20\%$) due to equilibrium limits; unreacted gas is recycled (4–5:1) after condensing product NH_3 .
- $>80\%$ of ammonia is used for fertilizers (urea, ammonium salts); emerging uses include a clean energy carrier and shipping fuel.

4.3 Nitric Acid: The Ostwald Process

- Three steps: (1) NH_3 oxidation on a Pt–Rh gauze catalyst, $\text{NH}_3 + \frac{5}{4}\text{O}_2 \longrightarrow \text{NO} + \frac{3}{2}\text{H}_2\text{O}$ (favoured at low P); (2) $2\text{NO} + \text{O}_2 \longrightarrow 2\text{NO}_2$ (favoured at high P, rate $\propto P^3$); (3) absorption, $3\text{NO}_2 + \text{H}_2\text{O} \longrightarrow 2\text{HNO}_3 + \text{NO}$.
- Plants may use single or dual pressure (oxidation at low P for higher conversion/lower catalyst loss, absorption at higher P); classified as low, medium or high pressure.
- Concentration above the $\text{HNO}_3\text{--H}_2\text{O}$ azeotrope (68 wt%) uses dehydrating agents such as strong H_2SO_4 or $\text{Mg}(\text{NO}_3)_2$ solution.
- Tail-gas NO_x is controlled by selective catalytic reduction with NH_3 (or adsorption on molecular sieves in older plants).

4.4 Chlor-Alkali Process

- Electrolysis of purified brine: anode $2\text{Cl}^- \longrightarrow \text{Cl}_2 + 2\text{e}^-$; cathode $2\text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{H}_2 + 2\text{OH}^-$; overall $2\text{NaCl} + 2\text{H}_2\text{O} \longrightarrow \text{Cl}_2 + \text{H}_2 + 2\text{NaOH}$.
- Cell technologies: mercury (toxic, phased out), diaphragm (lower purity), and membrane cells (ion-exchange membrane, highest efficiency and purity) – membrane is now standard.
- Electrodes: dimensionally stable Ti/RuO₂–TiO₂ anode; Ni-based cathode. Typical conditions: $70\text{--}90^\circ\text{C}$, 2–6 kA/m² current density, $\sim 25\text{--}30 \text{ wt}\%$ brine; energy use $\sim 2000\text{--}2600 \text{ kWh/ton Cl}_2$.
- Historical routes (Weldon: $\text{MnO}_2 + 4\text{HCl} \longrightarrow \text{Cl}_2 + \text{MnCl}_2 + 2\text{H}_2\text{O}$; Deacon: $4\text{HCl} + \text{O}_2 \rightleftharpoons 2\text{Cl}_2 + 2\text{H}_2\text{O}$) have been superseded by electrolysis. Chlorine feeds PVC, water treatment and organic/inorganic chlorination; caustic soda and H_2 are valuable co-products.

5 Fertilizer and Related Chemicals

5.1 Soda Ash: The Solvay Process

- Raw materials: brine (NaCl), limestone (CaCO_3) and ammonia (recycled catalyst/intermediate).
- Steps: (1) ammoniation of brine; (2) carbonation – $\text{NaCl} + \text{NH}_3 + \text{CO}_2 + \text{H}_2\text{O} \longrightarrow \text{NaHCO}_3 \downarrow + \text{NH}_4\text{Cl}$ (exothermic, cooled to $20\text{--}25^\circ\text{C}$ to maximise precipitation, since NaHCO_3 solubility falls with T); (3) calcination – $2\text{NaHCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$ ($150\text{--}230^\circ\text{C}$), regenerating CO_2 for recycle; (4) ammonia recovery – $2\text{NH}_4\text{Cl} + \text{Ca}(\text{OH})_2 \longrightarrow 2\text{NH}_3 + \text{CaCl}_2 + 2\text{H}_2\text{O}$, using lime from calcined limestone.
- Ammonia recycling and inexpensive raw materials make the process economical; it has replaced the older Leblanc process. Major use: glass manufacture (lowers silica melting point) and detergents.

5.2 Ammonium Sulphate

Produced by neutralising ammonia with sulfuric acid, $2\text{NH}_3 + \text{H}_2\text{SO}_4 \longrightarrow (\text{NH}_4)_2\text{SO}_4$ (exothermic; carried out in a saturator/reactor, no catalyst needed), followed by evaporation, crystallisation, centrifugation and drying. Reaction temperature $\sim 90\text{--}100^\circ\text{C}$; pH 4–5 controls purity; used mainly as a nitrogen–sulfur fertilizer, also obtained as a by-product of caprolactam production and coke-oven gas treatment.

5.3 Urea

- Two-step equilibrium synthesis: (1) $\text{CO}_2 + 2\text{NH}_3 \rightleftharpoons \text{NH}_2\text{COONH}_4$ (ammonium carbamate; highly exothermic, 180°C, 14 MPa); (2) $\text{NH}_2\text{COONH}_4 \rightleftharpoons \text{NH}_2\text{CONH}_2 + \text{H}_2\text{O}$ (carbamate dehydration; slow, endothermic).
- Stoichiometric conversion is only $\sim 55\%$; excess NH_3 (feed ratio $\text{NH}_3:\text{CO}_2 \approx 2.5\text{--}4:1$) shifts equilibrium and raises conversion to $\sim 85\%$ while also absorbing reaction heat and suppressing biuret formation.
- No conventional catalyst is used; the highly corrosive carbamate requires a small air bleed for passivation of construction materials. Unreacted NH_3/CO_2 are stripped and recycled (e.g. Stamicarbon process).
- Highest-N solid fertilizer ($\sim 46\%$ N); also used for melamine resins and diesel exhaust fluid (urea-SCR for NO_x control).

6 Industrial Gas Separation: Nitrogen Generation

- Nitrogen is separated from air (78% N_2 , 21% O_2) by exploiting differences in molecular size, adsorption affinity or diffusivity.
- **Pressure Swing Adsorption (PSA)**: a carbon molecular sieve kinetically adsorbs O_2 faster than N_2 under pressure; the bed is regenerated at low pressure (alternating dual beds for continuous output). Achieves very high purity ($>99.9\%$) with good adjustability.
- **Membrane separation**: O_2 , CO_2 and H_2O permeate a polymer membrane faster than N_2 (solution-diffusion mechanism); N_2 is retained as product. Simpler, continuous, lower cost, but limited to moderate purity ($\sim 95\text{--}99.5\%$).
- **Cryogenic distillation** is preferred for very large-scale, high-purity production, exploiting differences in boiling point, but is energy-intensive.

7 Methanol

- Chemistry: from syngas, $\text{CO} + 2\text{H}_2 \rightleftharpoons \text{CH}_3\text{OH}$ and $\text{CO}_2 + 3\text{H}_2 \rightleftharpoons \text{CH}_3\text{OH} + \text{H}_2\text{O}$ (both exothermic, reversible); catalyzed by $\text{Cu}/\text{ZnO}/\text{Al}_2\text{O}_3$ at 200–300°C, 50–100 atm.
- Syngas is generated by steam methane reforming ($\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3\text{H}_2$, $\sim 65\%$ of global production), coal gasification, or emerging biomass/ CO_2 -hydrogenation (e-methanol) routes.
- Since moles of gas decrease on reaction, higher pressure favours methanol formation (Le Chatelier); unreacted syngas is recycled with a purge to prevent inert build-up.
- Major uses: formaldehyde and acetic acid manufacture, fuel blending (M15/M85), biodiesel (transesterification), and MTBE production.

8 Hydrogen: Production, Storage and Utilization

- Hydrogen has a very high gravimetric energy density (~ 120 MJ/kg vs. ~ 44 MJ/kg for gasoline) but low volumetric density, motivating compressed, cryogenic or solid-sorbent storage.
- **Production**: steam reforming of hydrocarbons (endothermic; commercial for methane, LPG, naphtha, methanol) remains dominant (“grey” hydrogen, with CO_2 emissions); electrolysis of water ($\text{H}_2\text{O} \longrightarrow \text{H}_2 + \frac{1}{2}\text{O}_2$) using renewable electricity gives “green” hydrogen. PEM electrolyzers use a proton-exchange (Nafion-type) membrane, Pt (HER) and Ir/IrO₂ (OER) catalysts.
- **Utilization**: fuel cells run the reverse reaction, $\text{H}_2 + \frac{1}{2}\text{O}_2 \longrightarrow \text{H}_2\text{O}$, generating electricity with no direct emissions; cells are stacked in series/parallel for voltage/current.
- Refinery uses: hydrocracking and hydrotreating consume large volumes of H_2 . In fertilizers, H_2 from reforming or electrolysis feeds Haber–Bosch ammonia synthesis.
- India’s National Green Hydrogen Mission targets 5 MMT of green H_2 production by 2030 to offset import-dependent grey hydrogen.

9 Energy Storage: Batteries

- Motivation: renewable supply (solar, wind) is intermittent and low in energy density relative to demand, creating a supply–demand gap that storage helps bridge.
- A battery interconverts chemical and electrical energy via a redox couple at the cathode and anode; secondary

(rechargeable) batteries allow this interconversion to run in both directions.

- **Lithium-ion batteries** dominate portable/EV applications: Li is highly electronegative (high cell voltage), light, and its small ion size gives fast diffusion (high power). Cathode materials must reversibly intercalate large amounts of Li with high ionic/electronic conductivity and structural stability; electrolytes need high ionic but negligible electronic conductivity and chemical/thermal stability.
- Cell formats include cylindrical, prismatic, pouch and coin cells, each built from a cathode, anode, electrolyte and separator.

